What is a List in Python?

A **list** is an **ordered**, **changeable (mutable)** collection that can hold **multiple items** of **different data types**.

You can think of it as a **container** for data — like an array, but more flexible.

🧩 Creating a List

fruits = ["apple", "banana", "cherry"]

numbers = [1, 2, 3, 4, 5]

mixed = [10, "hello", 3.14, True]

empty = []

✅ Lists are enclosed in **square brackets [ ]**.
✅ Lists can store **different data types** in the same list.

🔹 Accessing List Items (Indexing)

Indexes start from **0**.

fruits = ["apple", "banana", "cherry"]

print(fruits[0]) # apple

print(fruits[1]) # banana

print(fruits[-1]) # cherry (last item)

🔹 Slicing Lists

You can extract parts (sub-lists) using slicing.

numbers = [10, 20, 30, 40, 50]

print(numbers[1:4]) # [20, 30, 40]

```
print(numbers[:3])    # [10, 20, 30]
print(numbers[2:])    # [30, 40, 50]
print(numbers[-3:-1]) # [30, 40]
```

🔹 Changing List Items

Lists are **mutable**, so you can modify them.

```
fruits = ["apple", "banana", "cherry"]
fruits[1] = "orange"
print(fruits)   # ['apple', 'orange', 'cherry']
```

🔹 Adding Items

Method	Description	Example
append()	Adds item at end	fruits.append("grape")
insert()	Adds item at specific index	fruits.insert(1, "mango")
extend()	Add elements of another list	fruits.extend(["kiwi", "melon"])

```
fruits = ["apple", "banana"]
fruits.append("cherry")
fruits.insert(1, "orange")
fruits.extend(["kiwi", "grape"])
print(fruits)
# ['apple', 'orange', 'banana', 'cherry', 'kiwi', 'grape']
```

🔹 Removing Items

Method	Description	Example
remove()	Removes by value	fruits.remove("banana")
pop()	Removes by index (default last)	fruits.pop(1)
del	Delete element by index	del fruits[0]
clear()	Remove all elements	fruits.clear()

```
fruits = ["apple", "banana", "cherry"]
fruits.remove("banana")
print(fruits)  # ['apple', 'cherry']
```

- **Loop Through a List**

```
fruits = ["apple", "banana", "cherry"]
for f in fruits:
  print(f)
```

Or using **index**:

```
for i in range(len(fruits)):
  print(fruits[i])
```

- **Check if Item Exists**

```
fruits = ["apple", "banana", "cherry"]
if "banana" in fruits:
  print("Yes, banana is in the list")
```

- **List Length**

```
fruits = ["apple", "banana", "cherry"]
print(len(fruits))  # 3
```

🔹 Copying Lists

Use .copy() or list() to copy.

```
fruits = ["apple", "banana", "cherry"]

new_list = fruits.copy()

print(new_list)
```

🔹 Joining Lists

You can **combine lists** using + or extend().

```
list1 = [1, 2, 3]

list2 = [4, 5]

result = list1 + list2

print(result)  # [1, 2, 3, 4, 5]
```

🔹 List Methods Summary

Method	Description
append()	Add element at end
insert()	Add at specific index
extend()	Merge another list
remove()	Remove by value
pop()	Remove by index
clear()	Remove all elements
sort()	Sort list
reverse()	Reverse order

Method	Description
copy()	Make a copy
count()	Count occurrences
index()	Return index of value

🧰 Example: Practical Usage

```
numbers = [5, 2, 9, 1, 5, 6]

numbers.sort()
print("Sorted:", numbers)

numbers.reverse()
print("Reversed:", numbers)

print("Count of 5:", numbers.count(5))
print("Max:", max(numbers))
print("Min:", min(numbers))
```

✅ Summary

Feature	Description	Example
Mutable	Can be changed	✅
Ordered	Items have fixed positions	✅
Duplicate Allowed	Yes	✅
Indexed	Access by index	✅

Feature	Description	Example
Mixed Data Types	Yes	✅